RF Exposure Evaluation Report

Product	:	Seeed Studio XIAO ESP32S3, Seeed Studio XIAO ESP32S3 Sense
Trade mark	:	Seeed Studio
Model/Type reference	:	XIAO-ESP32-S3, XIAO-ESP32-S3-Sense
Serial Number	:	N/A
Report Number	:	EED32P80440303
Date of Issue	:	Apr. 20, 2023
Test Standards	:	EN 50665:2017 EN IEC 62311:2020
Test result	:	PASS

Prepared for:

Seeed Technology Co., Ltd
9F, Building G3, TCL International E city, Zhongshanyuan Road, Nanshan, Shenzhen, China.

Prepared by:

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2 Version

Version No.	Date	Description
00	Apr. 20, 2023	Original

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4 General Information

4.1 Client Information

Applicant:	Seeed Technology Co., Ltd
Address of Applicant:	9F, Building G3, TCL International E city, Zhongshanyuan Road, Nanshan, Shenzhen, China.
Manufacturer:	Seeed Technology Co., Ltd
Address of Manufacturer:	9F, Building G3, TCL International E city, Zhongshanyuan Road, Nanshan, Shenzhen, China.
Factory:	Shenzhen Xinxian Technology Co; Limited
Address of Factory:	F5, Building B17, Hengfeng Industrial City, No. 739 Zhoushi Rd, Baoan District, Shenzhen, Guangdong, P.R.C.

4.2 General Description of EUT

Product Name:	Seeed Studio XIAO ESP32S3, Seeed Studio XIAO ESP32S3 Sense	
Model No.(EUT):	XIAO-ESP32-S3, XIAO-ESP32-S3-Sense	
Trade Mark:	Seeed Studio	
EUT Supports Radios application:	BLE, 2.4G WIFI	
Frequency Range:	BLE: 2402MHz to 2480MHz 2.4G WIFI: IEEE 802.11b/g/n(HT20): 2412MHz to 2472MHz IEEE 802.11n(HT40): 2422MHz to 2462MHz	
Modulation Type:	BLE: GFSK 2.4G WIFI: IEEE for 802.11b:DSSS(CCK,DQPSK,DBPSK) IEEE for 802.11g:OFDM(64QAM, 16QAM, QPSK, BPSK) IEEE for 802.11n(HT20 and HT40): OFDM (64QAM, 16QAM,QPSK,BPSK)	
Sample Type:	☐ Portable Use ☒ Fixed-Use ☐ Vehicle Use	
Test Power Grade:	Default	
Test Software of EUT:	ESP32S3.exe	
Antenna Type:	FPC antenna	
Antenna Gain:	2.90dBi	
Power Supply:	USB port:	DC 5.0V
Sample Received Date:	Mar. 31, 2023	
Sample tested Date:	Mar. 31, 2023 to Apr. 11, 2023	

Remark:
Company Name and Address shown on Report, the sample(s) and sample Information was/ were provided by the applicant who should be responsible for the authenticity which CTI hasn't verified.

Model No.: XIAO-ESP32-S3, XIAO-ESP32-S3-Sense
XIAO-ESP32-S3-Sense is composed of XIAO-ESP32-S3 main board and expansion board. XIAO-ESP32-S3 is a single main board without an expansion board, and compared with the above-mentioned XIAO-ESP32-S3 Sense main board, there is no connector on the board. Their electrical circuit design, layout, components used and internal wiring are identical.

4.3 Test Location

All tests were performed at:

Centre Testing International Group Co., Ltd.

Building C, Hongwei Industrial Park Block 70, Bao'an District, Shenzhen, China

Telephone: +86 (0) 755 3368 3668 Fax:+86 (0) 755 3368 3385

No tests were sub-contracted.

4.4 Deviation from Standards

None.

4.5 Abnormalities from Standard Conditions

None.

4.6 Other Information Requested by the Customer

None.

5 Technical Requirements Specification in EN 50665

5.1 General Description of Applied Standards

EN 50665 Generic standard to demonstrate the compliance of electronic and electrical apparatus with the basic restrictions related to human exposure to electromagnetic fields (0 Hz–300 GHz) is to demonstrate the compliance of apparatus with the basic restrictions or reference levels on exposure of the general public related to electric, magnetic, electromagnetic fields as well as induced and contact current

5.2 RF Exposure Evaluation

Limit

For equipment intended for use by the general public the relevant exposure restrictions in Council Recommendation 1999/519/EC shall be applied

Reference levels for electric, magnetic and electromagnetic fields (0 Hz to 300 GHz, unperturbed rms values)

Frequency range	E-field strength (V/m)	H-field strength (A/m)	B-field (μT)	Equivalent plane wave power density S_{eq} (W/m²)
0-1 Hz	—	$3,2 \times 10^4$	4×10^4	—
1-8 Hz	10 000	$3,2 \times 10^4/f^2$	$4 \times 10^4/f^2$	—
8-25 Hz	10 000	4 000/f	5 000/f	—
0,025-0,8 kHz	250/f	4/f	5/f	—
0,8-3 kHz	250/f	5	6,25	—
3-150 kHz	87	5	6,25	—
0,15-1 MHz	87	0,73/f	0,92/f	—
1-10 MHz	$87/f^{1/2}$	0,73/f	0,92/f	—
10-400 MHz	28	0,073	0,092	2
400-2 000 MHz	$1,375\ f^{1/2}$	$0,0037\ f^{1/2}$	$0,0046\ f^{1/2}$	f/200
2-300 GHz	61	0,16	0,20	10

Notes:

1. f as indicated in the frequency range column.
2. For frequencies between 100 kHz and 10 GHz, S_{eq}, E^2, H^2, and B^2 are to be averaged over any six-minute period.
3. For frequencies exceeding 10 GHz, S_{eq}, E^2, H^2, and B^2 are to be averaged over any $68/f^{1,05}$ -minute period (f in GHz).
4. No E-field value is provided for frequencies < 1 Hz, which are effectively static electric fields. For most people the annoying perception of surface electric charges will not occur at field strengths less than 25 kV/m. Spark discharges causing stress or annoyance should be avoided.

5.2.1Human Exposure Assessment

Exposure evaluation	
Given $E = \frac{\sqrt{30 \times TP}}{D}$ $D = \frac{\sqrt{30 \times TP}}{E}$	Where: ● E: E field Strength ● TP: Transmitted power in watt ● D: Distance from the transmitting antenna in meter

For Stand alone:

Bluetooth LE:

Frequency	EIRP (dBm)	TP (W)	D (m)	Electric Field(V/m)	Limit of Electric Field (V/m)	Result
2480	2.12	0.0016	0.2	1.11	61	Pass

2.4G WIFI:

Frequency	EIRP (dBm)	TP (W)	D (m)	Electric Field(V/m)	Limit of Electric Field (V/m)	Result
2472	16.81	0.0480	0.2	6.00	61	Pass

Conclusion:

→*E= 6.00V/m (max) is the E-Field strength when safety distance between the EUT and human body is 0.2m, which is below 61V/m as required 1999/519/EC Annex III Table 2*

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*** End of Report ***